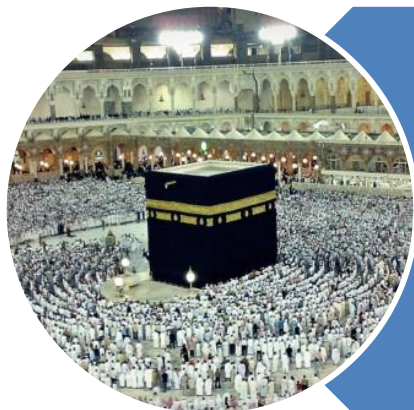


رَبِّ اشْرَحْ لِي صَدْرِي وَيَسِّرْ لِي أَمْرِي وَاحْلُلْ عُقْدَةً مِنْ لِسَانِي يَفْقَهُوا قَوْلِي



O my Lord! Expand for me my chest [with assurance] and ease for me my task and untie the knot from my tongue that they may understand my speech. **(Quran 20 : 25-28)**

# Quadratic Sorting

## Bubble Sort

C

P

C

S

2

0

4

3

## Bubble Sort - **Algorithm**

- You always compare consecutive elements
  - Going left to right
- Whenever two elements are out of place,
  - SWAP them
- At the end of a single iteration,
  - the maximum element will be in the last spot
- Now you simply repeat this n times
  - where n is the number of elements being sorted

## Bubble Sort

| Index   | 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  |
|---------|----|----|----|----|----|----|----|----|
| Initial | 23 | 77 | 45 | 64 | 28 | 35 | 18 | 40 |
|         |    |    |    |    |    |    |    |    |
|         |    |    |    |    |    |    |    |    |
|         |    |    |    |    |    |    |    |    |
|         |    |    |    |    |    |    |    |    |
|         |    |    |    |    |    |    |    |    |
|         |    |    |    |    |    |    |    |    |
|         |    |    |    |    |    |    |    |    |
| Final   |    |    |    |    |    |    |    |    |

## Bubble Sort

| Index       | 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  |
|-------------|----|----|----|----|----|----|----|----|
| Initial     | 23 | 77 | 45 | 64 | 28 | 35 | 18 | 40 |
| Iteration 1 | 23 | 77 | 45 | 64 | 28 | 35 | 18 | 40 |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
| Final       |    |    |    |    |    |    |    |    |

## Bubble Sort

| Index       | 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  |
|-------------|----|----|----|----|----|----|----|----|
| Initial     | 23 | 77 | 45 | 64 | 28 | 35 | 18 | 40 |
| Iteration 1 | 23 | 77 | 45 | 64 | 28 | 35 | 18 | 40 |
|             | 23 | 45 | 77 | 64 | 28 | 35 | 18 | 40 |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
| Final       |    |    |    |    |    |    |    |    |

## Bubble Sort

| Index       | 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  |
|-------------|----|----|----|----|----|----|----|----|
| Initial     | 23 | 77 | 45 | 64 | 28 | 35 | 18 | 40 |
| Iteration 1 | 23 | 77 | 45 | 64 | 28 | 35 | 18 | 40 |
|             | 23 | 45 | 77 | 64 | 28 | 35 | 18 | 40 |
|             | 23 | 45 | 64 | 77 | 28 | 35 | 18 | 40 |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
| Final       |    |    |    |    |    |    |    |    |

## Bubble Sort

| Index       | 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  |
|-------------|----|----|----|----|----|----|----|----|
| Initial     | 23 | 77 | 45 | 64 | 28 | 35 | 18 | 40 |
| Iteration 1 | 23 | 77 | 45 | 64 | 28 | 35 | 18 | 40 |
|             | 23 | 45 | 77 | 64 | 28 | 35 | 18 | 40 |
|             | 23 | 45 | 64 | 77 | 28 | 35 | 18 | 40 |
|             | 23 | 45 | 64 | 28 | 77 | 35 | 18 | 40 |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
| Final       |    |    |    |    |    |    |    |    |



## Bubble Sort

| Index       | 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  |
|-------------|----|----|----|----|----|----|----|----|
| Initial     | 23 | 77 | 45 | 64 | 28 | 35 | 18 | 40 |
| Iteration 1 | 23 | 77 | 45 | 64 | 28 | 35 | 18 | 40 |
|             | 23 | 45 | 77 | 64 | 28 | 35 | 18 | 40 |
|             | 23 | 45 | 64 | 77 | 28 | 35 | 18 | 40 |
|             | 23 | 45 | 64 | 28 | 77 | 35 | 18 | 40 |
|             | 23 | 45 | 64 | 28 | 35 | 77 | 18 | 40 |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
| Final       |    |    |    |    |    |    |    |    |

## Bubble Sort

| Index       | 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  |
|-------------|----|----|----|----|----|----|----|----|
| Initial     | 23 | 77 | 45 | 64 | 28 | 35 | 18 | 40 |
| Iteration 1 | 23 | 77 | 45 | 64 | 28 | 35 | 18 | 40 |
|             | 23 | 45 | 77 | 64 | 28 | 35 | 18 | 40 |
|             | 23 | 45 | 64 | 77 | 28 | 35 | 18 | 40 |
|             | 23 | 45 | 64 | 28 | 77 | 35 | 18 | 40 |
|             | 23 | 45 | 64 | 28 | 35 | 77 | 18 | 40 |
|             | 23 | 45 | 64 | 28 | 35 | 18 | 77 | 40 |
|             |    |    |    |    |    |    |    |    |
| Final       |    |    |    |    |    |    |    |    |

## Bubble Sort

| Index       | 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  |
|-------------|----|----|----|----|----|----|----|----|
| Initial     | 23 | 77 | 45 | 64 | 28 | 35 | 18 | 40 |
| Iteration 1 | 23 | 77 | 45 | 64 | 28 | 35 | 18 | 40 |
|             | 23 | 45 | 77 | 64 | 28 | 35 | 18 | 40 |
|             | 23 | 45 | 64 | 77 | 28 | 35 | 18 | 40 |
|             | 23 | 45 | 64 | 28 | 77 | 35 | 18 | 40 |
|             | 23 | 45 | 64 | 28 | 35 | 77 | 18 | 40 |
|             | 23 | 45 | 64 | 28 | 35 | 18 | 77 | 40 |
|             | 23 | 45 | 64 | 28 | 35 | 18 | 40 | 77 |
| Final       |    |    |    |    |    |    |    |    |

## Bubble Sort

| Index       | 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  |
|-------------|----|----|----|----|----|----|----|----|
| Initial     | 23 | 77 | 45 | 64 | 28 | 35 | 18 | 40 |
| Iteration 2 | 23 | 45 | 64 | 28 | 35 | 18 | 40 | 77 |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
| Final       |    |    |    |    |    |    |    |    |

## Bubble Sort

| Index       | 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  |
|-------------|----|----|----|----|----|----|----|----|
| Initial     | 23 | 77 | 45 | 64 | 28 | 35 | 18 | 40 |
| Iteration 2 | 23 | 45 | 64 | 28 | 35 | 18 | 40 | 77 |
|             | 23 | 45 | 64 | 28 | 35 | 18 | 40 | 77 |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
| Final       |    |    |    |    |    |    |    |    |

## Bubble Sort

| Index       | 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  |
|-------------|----|----|----|----|----|----|----|----|
| Initial     | 23 | 77 | 45 | 64 | 28 | 35 | 18 | 40 |
| Iteration 2 | 23 | 45 | 64 | 28 | 35 | 18 | 40 | 77 |
|             | 23 | 45 | 64 | 28 | 35 | 18 | 40 | 77 |
|             | 23 | 45 | 64 | 28 | 35 | 18 | 40 | 77 |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
| Final       |    |    |    |    |    |    |    |    |

## Bubble Sort

| Index       | 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  |
|-------------|----|----|----|----|----|----|----|----|
| Initial     | 23 | 77 | 45 | 64 | 28 | 35 | 18 | 40 |
| Iteration 2 | 23 | 45 | 64 | 28 | 35 | 18 | 40 | 77 |
|             | 23 | 45 | 64 | 28 | 35 | 18 | 40 | 77 |
|             | 23 | 45 | 64 | 28 | 35 | 18 | 40 | 77 |
|             | 23 | 45 | 28 | 64 | 35 | 18 | 40 | 77 |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
| Final       |    |    |    |    |    |    |    |    |

## Bubble Sort

| Index       | 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  |
|-------------|----|----|----|----|----|----|----|----|
| Initial     | 23 | 77 | 45 | 64 | 28 | 35 | 18 | 40 |
| Iteration 2 | 23 | 45 | 64 | 28 | 35 | 18 | 40 | 77 |
|             | 23 | 45 | 64 | 28 | 35 | 18 | 40 | 77 |
|             | 23 | 45 | 64 | 28 | 35 | 18 | 40 | 77 |
|             | 23 | 45 | 28 | 64 | 35 | 18 | 40 | 77 |
|             | 23 | 45 | 28 | 35 | 64 | 18 | 40 | 77 |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
| Final       |    |    |    |    |    |    |    |    |



## Bubble Sort

| Index       | 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  |
|-------------|----|----|----|----|----|----|----|----|
| Initial     | 23 | 77 | 45 | 64 | 28 | 35 | 18 | 40 |
| Iteration 2 | 23 | 45 | 64 | 28 | 35 | 18 | 40 | 77 |
|             | 23 | 45 | 64 | 28 | 35 | 18 | 40 | 77 |
|             | 23 | 45 | 64 | 28 | 35 | 18 | 40 | 77 |
|             | 23 | 45 | 28 | 64 | 35 | 18 | 40 | 77 |
|             | 23 | 45 | 28 | 35 | 64 | 18 | 40 | 77 |
|             | 23 | 45 | 28 | 35 | 18 | 64 | 40 | 77 |
|             |    |    |    |    |    |    |    |    |
| Final       |    |    |    |    |    |    |    |    |

## Bubble Sort

| Index       | 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  |
|-------------|----|----|----|----|----|----|----|----|
| Initial     | 23 | 77 | 45 | 64 | 28 | 35 | 18 | 40 |
| Iteration 2 | 23 | 45 | 64 | 28 | 35 | 18 | 40 | 77 |
|             | 23 | 45 | 64 | 28 | 35 | 18 | 40 | 77 |
|             | 23 | 45 | 64 | 28 | 35 | 18 | 40 | 77 |
|             | 23 | 45 | 28 | 64 | 35 | 18 | 40 | 77 |
|             | 23 | 45 | 28 | 35 | 64 | 18 | 40 | 77 |
|             | 23 | 45 | 28 | 35 | 18 | 64 | 40 | 77 |
|             | 23 | 45 | 28 | 35 | 18 | 40 | 64 | 77 |
| Final       |    |    |    |    |    |    |    |    |

## Bubble Sort

| Index       | 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  |
|-------------|----|----|----|----|----|----|----|----|
| Initial     | 23 | 77 | 45 | 64 | 28 | 35 | 18 | 40 |
| Iteration 3 | 23 | 45 | 28 | 35 | 18 | 40 | 64 | 77 |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
| Final       |    |    |    |    |    |    |    |    |

## Bubble Sort

| Index       | 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  |
|-------------|----|----|----|----|----|----|----|----|
| Initial     | 23 | 77 | 45 | 64 | 28 | 35 | 18 | 40 |
| Iteration 3 | 23 | 45 | 28 | 35 | 18 | 40 | 64 | 77 |
|             | 23 | 45 | 28 | 35 | 18 | 40 | 64 | 77 |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
| Final       |    |    |    |    |    |    |    |    |

## Bubble Sort

| Index       | 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  |
|-------------|----|----|----|----|----|----|----|----|
| Initial     | 23 | 77 | 45 | 64 | 28 | 35 | 18 | 40 |
| Iteration 3 | 23 | 45 | 28 | 35 | 18 | 40 | 64 | 77 |
|             | 23 | 45 | 28 | 35 | 18 | 40 | 64 | 77 |
|             | 23 | 28 | 45 | 35 | 18 | 40 | 64 | 77 |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
| Final       |    |    |    |    |    |    |    |    |

## Bubble Sort

| Index       | 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  |
|-------------|----|----|----|----|----|----|----|----|
| Initial     | 23 | 77 | 45 | 64 | 28 | 35 | 18 | 40 |
| Iteration 3 | 23 | 45 | 28 | 35 | 18 | 40 | 64 | 77 |
|             | 23 | 45 | 28 | 35 | 18 | 40 | 64 | 77 |
|             | 23 | 28 | 45 | 35 | 18 | 40 | 64 | 77 |
|             | 23 | 28 | 35 | 45 | 18 | 40 | 64 | 77 |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
| Final       |    |    |    |    |    |    |    |    |

## Bubble Sort

| Index       | 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  |
|-------------|----|----|----|----|----|----|----|----|
| Initial     | 23 | 77 | 45 | 64 | 28 | 35 | 18 | 40 |
| Iteration 3 | 23 | 45 | 28 | 35 | 18 | 40 | 64 | 77 |
|             | 23 | 45 | 28 | 35 | 18 | 40 | 64 | 77 |
|             | 23 | 28 | 45 | 35 | 18 | 40 | 64 | 77 |
|             | 23 | 28 | 35 | 45 | 18 | 40 | 64 | 77 |
|             | 23 | 28 | 35 | 18 | 45 | 40 | 64 | 77 |
|             |    |    |    |    |    |    |    |    |
| Final       |    |    |    |    |    |    |    |    |

## Bubble Sort

| Index       | 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  |
|-------------|----|----|----|----|----|----|----|----|
| Initial     | 23 | 77 | 45 | 64 | 28 | 35 | 18 | 40 |
| Iteration 3 | 23 | 45 | 28 | 35 | 18 | 40 | 64 | 77 |
|             | 23 | 45 | 28 | 35 | 18 | 40 | 64 | 77 |
|             | 23 | 28 | 45 | 35 | 18 | 40 | 64 | 77 |
|             | 23 | 28 | 35 | 45 | 18 | 40 | 64 | 77 |
|             | 23 | 28 | 35 | 18 | 45 | 40 | 64 | 77 |
|             | 23 | 28 | 35 | 18 | 40 | 45 | 64 | 77 |
| Final       |    |    |    |    |    |    |    |    |



## Bubble Sort

| Index       | 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  |
|-------------|----|----|----|----|----|----|----|----|
| Initial     | 23 | 77 | 45 | 64 | 28 | 35 | 18 | 40 |
| Iteration 4 | 23 | 28 | 35 | 18 | 40 | 45 | 64 | 77 |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
| Final       |    |    |    |    |    |    |    |    |

## Bubble Sort

| Index       | 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  |
|-------------|----|----|----|----|----|----|----|----|
| Initial     | 23 | 77 | 45 | 64 | 28 | 35 | 18 | 40 |
| Iteration 4 | 23 | 28 | 35 | 18 | 40 | 45 | 64 | 77 |
|             | 23 | 28 | 35 | 18 | 40 | 45 | 64 | 77 |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
| Final       |    |    |    |    |    |    |    |    |

## Bubble Sort

| Index       | 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  |
|-------------|----|----|----|----|----|----|----|----|
| Initial     | 23 | 77 | 45 | 64 | 28 | 35 | 18 | 40 |
| Iteration 4 | 23 | 28 | 35 | 18 | 40 | 45 | 64 | 77 |
|             | 23 | 28 | 35 | 18 | 40 | 45 | 64 | 77 |
|             | 23 | 28 | 35 | 18 | 40 | 45 | 64 | 77 |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
| Final       |    |    |    |    |    |    |    |    |

## Bubble Sort

| Index       | 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  |
|-------------|----|----|----|----|----|----|----|----|
| Initial     | 23 | 77 | 45 | 64 | 28 | 35 | 18 | 40 |
| Iteration 4 | 23 | 28 | 35 | 18 | 40 | 45 | 64 | 77 |
|             | 23 | 28 | 35 | 18 | 40 | 45 | 64 | 77 |
|             | 23 | 28 | 35 | 18 | 40 | 45 | 64 | 77 |
|             | 23 | 28 | 18 | 35 | 40 | 45 | 64 | 77 |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
| Final       |    |    |    |    |    |    |    |    |

## Bubble Sort

| Index       | 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  |
|-------------|----|----|----|----|----|----|----|----|
| Initial     | 23 | 77 | 45 | 64 | 28 | 35 | 18 | 40 |
| Iteration 4 | 23 | 28 | 35 | 18 | 40 | 45 | 64 | 77 |
|             | 23 | 28 | 35 | 18 | 40 | 45 | 64 | 77 |
|             | 23 | 28 | 35 | 18 | 40 | 45 | 64 | 77 |
|             | 23 | 28 | 18 | 35 | 40 | 45 | 64 | 77 |
|             | 23 | 28 | 18 | 35 | 40 | 45 | 64 | 77 |
|             |    |    |    |    |    |    |    |    |
| Final       |    |    |    |    |    |    |    |    |

## Bubble Sort

| Index       | 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  |
|-------------|----|----|----|----|----|----|----|----|
| Initial     | 23 | 77 | 45 | 64 | 28 | 35 | 18 | 40 |
| Iteration 5 | 23 | 28 | 18 | 35 | 40 | 45 | 64 | 77 |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
| Final       |    |    |    |    |    |    |    |    |

## Bubble Sort

| Index       | 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  |
|-------------|----|----|----|----|----|----|----|----|
| Initial     | 23 | 77 | 45 | 64 | 28 | 35 | 18 | 40 |
| Iteration 5 | 23 | 28 | 18 | 35 | 40 | 45 | 64 | 77 |
|             | 23 | 28 | 18 | 35 | 40 | 45 | 64 | 77 |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
| Final       |    |    |    |    |    |    |    |    |

## Bubble Sort

| Index       | 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  |
|-------------|----|----|----|----|----|----|----|----|
| Initial     | 23 | 77 | 45 | 64 | 28 | 35 | 18 | 40 |
| Iteration 5 | 23 | 28 | 18 | 35 | 40 | 45 | 64 | 77 |
|             | 23 | 28 | 18 | 35 | 40 | 45 | 64 | 77 |
|             | 23 | 18 | 28 | 35 | 40 | 45 | 64 | 77 |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
| Final       |    |    |    |    |    |    |    |    |



## Bubble Sort

| Index       | 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  |
|-------------|----|----|----|----|----|----|----|----|
| Initial     | 23 | 77 | 45 | 64 | 28 | 35 | 18 | 40 |
| Iteration 5 | 23 | 28 | 18 | 35 | 40 | 45 | 64 | 77 |
|             | 23 | 28 | 18 | 35 | 40 | 45 | 64 | 77 |
|             | 23 | 18 | 28 | 35 | 40 | 45 | 64 | 77 |
|             | 23 | 18 | 28 | 35 | 40 | 45 | 64 | 77 |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
| Final       |    |    |    |    |    |    |    |    |

## Bubble Sort

| Index       | 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  |
|-------------|----|----|----|----|----|----|----|----|
| Initial     | 23 | 77 | 45 | 64 | 28 | 35 | 18 | 40 |
| Iteration 6 | 23 | 18 | 28 | 35 | 40 | 45 | 64 | 77 |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
| Final       |    |    |    |    |    |    |    |    |

## Bubble Sort

| Index       | 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  |
|-------------|----|----|----|----|----|----|----|----|
| Initial     | 23 | 77 | 45 | 64 | 28 | 35 | 18 | 40 |
| Iteration 6 | 23 | 18 | 28 | 35 | 40 | 45 | 64 | 77 |
|             | 18 | 23 | 28 | 35 | 40 | 45 | 64 | 77 |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
| Final       |    |    |    |    |    |    |    |    |

## Bubble Sort

| Index       | 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  |
|-------------|----|----|----|----|----|----|----|----|
| Initial     | 23 | 77 | 45 | 64 | 28 | 35 | 18 | 40 |
| Iteration 6 | 23 | 18 | 28 | 35 | 40 | 45 | 64 | 77 |
|             | 18 | 23 | 28 | 35 | 40 | 45 | 64 | 77 |
|             | 18 | 23 | 28 | 35 | 40 | 45 | 64 | 77 |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
| Final       |    |    |    |    |    |    |    |    |

## Bubble Sort

| Index       | 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  |
|-------------|----|----|----|----|----|----|----|----|
| Initial     | 23 | 77 | 45 | 64 | 28 | 35 | 18 | 40 |
| Iteration 7 | 18 | 23 | 28 | 35 | 40 | 45 | 64 | 77 |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
| Final       |    |    |    |    |    |    |    |    |

## Bubble Sort

| Index       | 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  |
|-------------|----|----|----|----|----|----|----|----|
| Initial     | 23 | 77 | 45 | 64 | 28 | 35 | 18 | 40 |
| Iteration 7 | 18 | 23 | 28 | 35 | 40 | 45 | 64 | 77 |
|             | 18 | 23 | 28 | 35 | 40 | 45 | 64 | 77 |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
| Final       |    |    |    |    |    |    |    |    |

## Bubble Sort

| Index       | 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  |
|-------------|----|----|----|----|----|----|----|----|
| Initial     | 23 | 77 | 45 | 64 | 28 | 35 | 18 | 40 |
| Iteration 8 | 18 | 23 | 28 | 35 | 40 | 45 | 64 | 77 |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
| Final       |    |    |    |    |    |    |    |    |

## Bubble Sort

| Index       | 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  |
|-------------|----|----|----|----|----|----|----|----|
| Initial     | 23 | 77 | 45 | 64 | 28 | 35 | 18 | 40 |
| Iteration 8 | 18 | 23 | 28 | 35 | 40 | 45 | 64 | 77 |
|             | 18 | 23 | 28 | 35 | 40 | 45 | 64 | 77 |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
| Final       |    |    |    |    |    |    |    |    |



## Bubble Sort

| Index       | 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  |
|-------------|----|----|----|----|----|----|----|----|
| Initial     | 23 | 77 | 45 | 64 | 28 | 35 | 18 | 40 |
| Iteration 8 | 18 | 23 | 28 | 35 | 40 | 45 | 64 | 77 |
|             | 18 | 23 | 28 | 35 | 40 | 45 | 64 | 77 |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
|             |    |    |    |    |    |    |    |    |
| Final       | 18 | 23 | 28 | 35 | 40 | 45 | 64 | 77 |

C

P

C

S

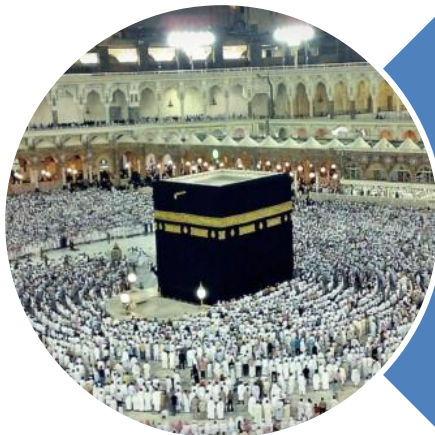
2

0

4

## Bubble Sort - Analysis

- $T(n) = n + (n-1) + (n-2) + (n-3) + \dots + 3 + 2 + 1$
- $T(n) = n(n+1)/2$
- $T(n) = \frac{1}{2}n^2 + \frac{1}{2}n$
- $T(n) = O(n^2)$



Allah (Alone) is Sufficient for us,  
and He is the Best Disposer of  
affairs (for us). **(Quran 3 : 173)**